

camera for tram

camera choice

AR0234 Global Shutter 1080P GigE Camera



Feature	Specification
Resolution	2.3 Megapixels (1920 × 1200)
Shutter Speed	Global Shutter (10μs up to 33ms)
FPS	60 fps at 1920×1200 / 1080p, 120 fps at 720p / VGA
Field of View (FOV)	80° to 105° (Diagonal), or around 85° Horizontal & 69° Vertical.
Zoom In/Out Capability	no
Power supply	PoE
Distance Coverage	Typically 2 meters to infinity by default, depending heavily on the focal length of the attached M12 lens.
Price	3589.1 HKD
Interface	GigE (Gigabit Ethernet)
Dimension	38mm x 38mm
Weight	25g

Feature	Specification
Network Interface	Gigabit Ethernet (RJ45), PoE (802.3af), GPIO header
Video Streaming Protocol	GigE Vision, ONVIF Profile S/T/G/M, H.264/H.265/MJPEG, RTSP
Control Interface	VISPA NXT SDK (C/C++/Python), GenICam, HW/SW trigger via GPIO, desktop app (Win/Linux)
Band Width	1 Gbps (1000Base-T)
Lens Mount	M12 S-Mount (interchangeable)

AR0234 Color Global Shutter MIPI Camera



Feature	Specification
Resolution	2.3 Megapixels (1920 × 1200)
Shutter Speed	Global Shutter (10μs to 3000ms)
FPS	60 fps at 1920×1200 (2-lane MIPI with ISP), up to 120 fps at lower resolutions
Field of View (FOV)	3.6mm M12 lens: 90° Horizontal, 75° Vertical, 120° Diagonal
Zoom In/Out Capability	no
Power supply	DC power
Distance Coverage	Focus range from 20cm to infinity
Price	1541.51 HKD
Interface	2 Lane MIPI CSI-2 & 4 Lane MIPI CSI-2

Feature	Specification
Dimension	38mm x 38mm
Weight	25g
Network Interface	MIPI CSI-2 (15-pin FFC ribbon, direct to SoC) — no Ethernet/Wi-Fi, requires host board
Video Streaming Protocol	RAW Bayer via V4L2; GStreamer → UDP/RTSP via host (no onboard streaming)
Control Interface	I²C, V4L2 ioctl, C++/Python SDK, HW trigger pin
Band Width	2-lane MIPI: 1.8 Gbps; 4-lane MIPI: 3.6 Gbps
Lens Mount	M12 S-Mount (interchangeable)

AR0234 Global Shutter 1080P WIFI Camera



Feature	Specification
Resolution	2.3 Megapixels (Active Array: 1920 × 1200 or 1080P mode)
Shutter Speed	Global Shutter (Eliminates high-speed rolling artifacts/jello effect)
FPS	30 fps at 1080p (WiFi bandwidth constrained), up to 60 fps at 720p
Field of View (FOV)	105° Diagonal (DFOV)
Zoom In/Out Capability	No
Power supply	PoE
Distance Coverage	Focus range from 20cm to infinity
Price	4567.85 HKD
Interface	Wi-Fi 802.11ac / Gigabit Ethernet (PoE)

Feature	Specification
Dimension	38mm x 38mm
Weight	25g
Network Interface	Dual-Band Wi-Fi 802.11ac (2.4/5 GHz) + RJ45 PoE (802.3af), BT 5.0
Video Streaming Protocol	RTSP, ONVIF Profile S/T, H.264/H.265/MJPEG
Control Interface	Vadzo360 desktop app (Win/Linux), ONVIF, libcamera, HW trigger via GPIO
Band Width	Wi-Fi up to 433 Mbps; GigE PoE 1 Gbps
Lens Mount	M12 S-Mount (interchangeable)

MV-CH100-60UC

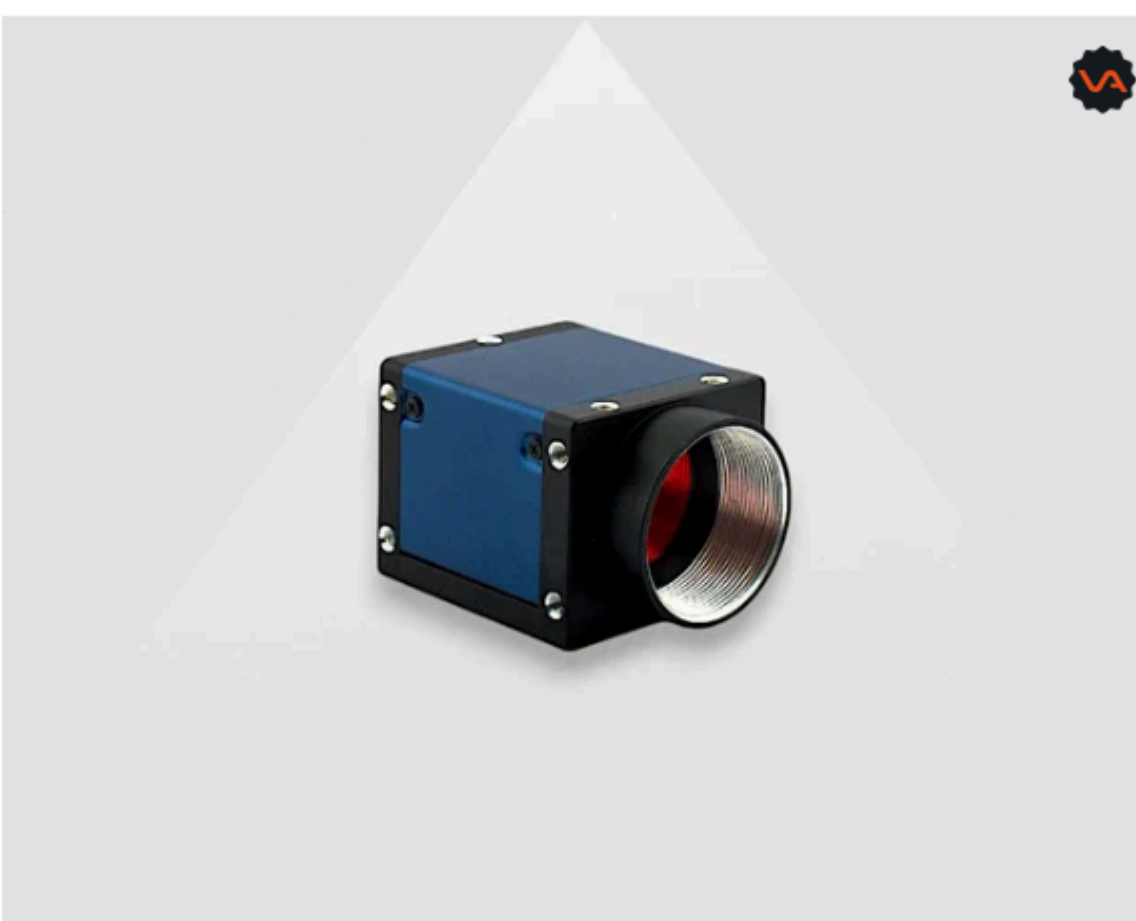


- Hikrobot 1/1.8", 12mm, f2.4, 10MP, C-Mount Lens
 - Φ 29 × 37 mm

Feature	Specification
Resolution	10 Megapixels (4096 × 2460)
Shutter Speed	Global Shutter (80μs to 10 seconds)
FPS	36 fps at 4096×2460
Field of View (FOV)	Lens Dependent

Feature	Specification
Zoom In/Out Capability	No
Power supply	PoE
Distance Coverage	Adjustable to infinity
Price	~3000 HKD
Interface	USB3.0, 機器視覺全能型智能相機客戶端SCVM (Machine Vision All-in-One Smart Camera Client SCVM)
Dimension	29 mm × 44 mm × 59 mm
Weight	113 g
Network Interface	USB 3.0 Micro-B (no Ethernet, no Wi-Fi)
Video Streaming Protocol	USB3 Vision (no RTSP — host-side streaming only via SDK)
Control Interface	MVS desktop app, GenICam, HW/SW trigger via 6-pin P7 GPIO
Band Width	USB 3.0 (5 Gbps)
Lens Mount	C-Mount

Daheng Imaging ME2P-1230-23U3M



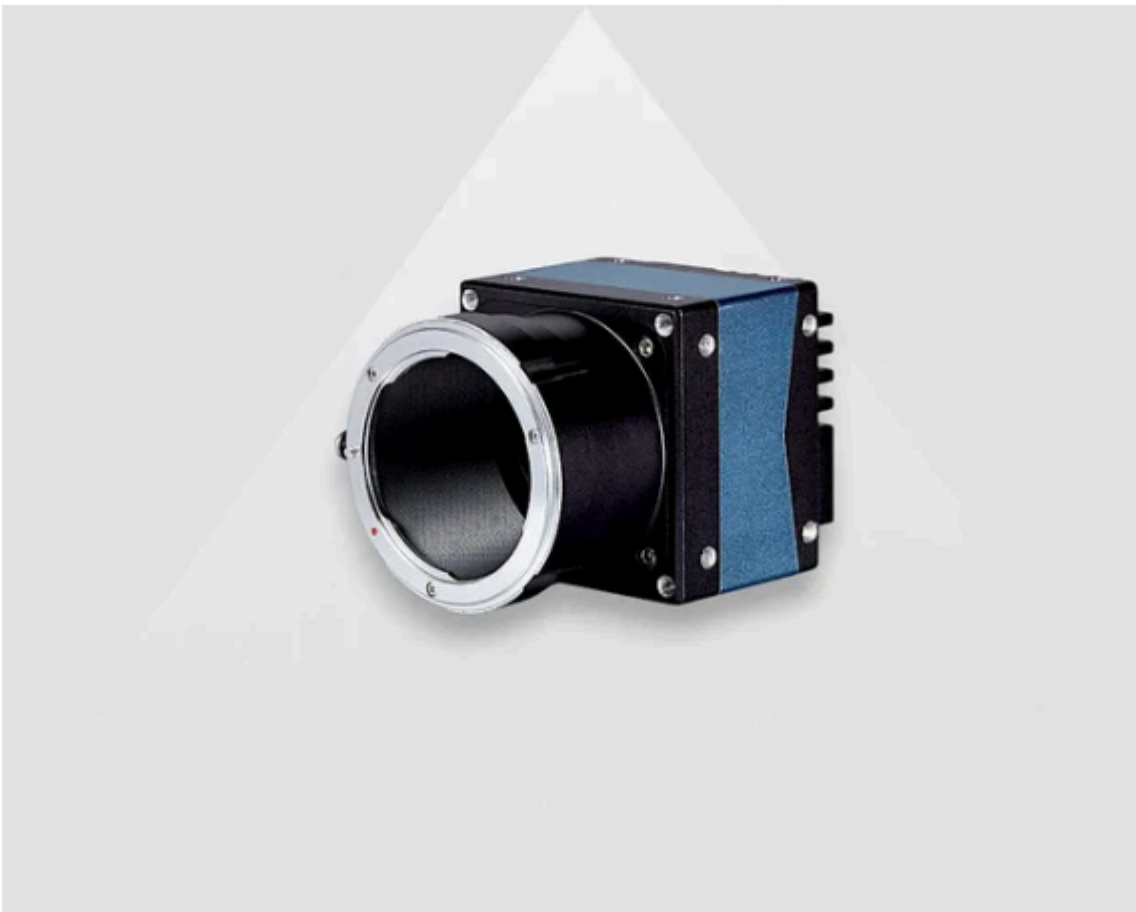
- VA-LCM-20MP-50MM-F2.8-110 Lens C-mount
 - Φ 50.5 × 53.9 mm

Feature	Specification
Resolution	12 Megapixels (4096 × 3000)
Shutter Speed	Global Shutter (28μs to 1s)
FPS	23.5 fps at 4096×3000
Field of View (FOV)	lens dependance
Zoom In/Out Capability	No
Power supply	USB
Distance Coverage	lens dependance
Price	€1.127,00
Interface	USB3 Vision Galaxy_Linux-x86_Gige-U3_32bits-64bits_2.4.2507.9231 SDK
Dimension	36mm × 31mm × 38.8mm
Weight	66g
Network Interface	USB 3.0 Micro-B (locking screws), no Ethernet
Video Streaming Protocol	USB3 Vision (no RTSP — host-side streaming only via SDK)
Control Interface	Daheng SDK (C++/Python, Win/Linux/ARM), GenICam, HW/SW trigger via GPIO
Band Width	USB 3.0 (5 Gbps)
Lens Mount	C-Mount / CS-Mount

Daheng Imaging MARS-3140-3GM/C-P

F mount

- AF-S DX MICRO NIKKOR 85MM F/3.5G ED VR
 - Φ 73.0 × 98.5 mm



Feature	Specification
Resolution	31.4 Megapixels (6464 × 4852)
Shutter Speed	Global Shutter (63 μs ~ 1 s)
FPS	3.4 fps at 6464×4852
Field of View (FOV)	lens dependance
Zoom In/Out Capability	No
Power supply	PoE
Distance Coverage	lens dependance
Price	¥ 40,800 CNY
Interface	GigE (Gigabit Ethernet) with Galaxy_Linux-x86_Gige-U3_32bits-64bits_2.4.2507.9231 SDK
Dimension	52.1mm × 62mm × 62mm
Weight	292g
Network Interface	Gigabit Ethernet (RJ45, locking screws), PoE (802.3af), 12-pin Hirose GPIO
Video Streaming Protocol	GigE Vision (no RTSP — industrial protocol via SDK)

Feature	Specification
Control Interface	Daheng SDK (C++/Python, Win/Linux/ARM/Mac), GenICam, HW/SW trigger via 12-pin Hirose GPIO
Band Width	1 Gbps (1000Base-T)
Lens Mount	F-Mount (default) or M42-Mount

Teledyne FLIR Blackfly S USB3 (BFS-U3-88S6M-C)



lens

- Teledyne FLIR Lenses
 - 8mm: Φ 70.4 × 90.5 mm
 - 12mm: Φ 70.4 × 90.5 mm
 - 16mm: Φ 48.0 × 58.0 – 60.4 mm
 - 25mm: Φ 37.5 × 45.3 mm
 - 35mm: Φ 48.1 × 65.8 mm
 - 50mm: Φ 50.0 × 76.6 mm

Feature	Specification
Resolution	8.9 Megapixels (4096 × 2160)
Shutter Speed	Global Shutter (10 μs to 30 seconds)
FPS	32 fps at 4096×2160 (Mono8), 26 fps (Mono12p), 16 fps (Mono16)
Field of View (FOV)	Lens dependance
Zoom In/Out Capability	No
Power supply	USB
Distance Coverage	Lens dependance
Price	~4,800 HKD
Interface	USB 3.1

Feature	Specification
Dimension	29 mm × 29 mm × 30
Weight	36g
Network Interface	USB 3.1 Gen 1 Micro-B (locking screws), Spinnaker SDK
Video Streaming Protocol	USB3 Vision (no RTSP — host-side streaming only via SDK)
Control Interface	Spinnaker SDK (C/C++/C#/VB.NET), SpinView desktop app, GenICam, HW/SW trigger via 6-pin Hirose GPIO
Band Width	USB 3.1 Gen 1 (5 Gbps)
Lens Mount	C-Mount

Basler ace 2 R Basic (a2A4096-30umBAS)



Feature	Specification
Resolution	12.3 Megapixels (4096 × 3000)
Shutter Speed	Global Shutter (14μs to 10 seconds)
FPS	29.3 fps at 4096×3000 (default), up to 31.7 fps (throughput limit off)
Field of View (FOV)	Lens dependance
Zoom In/Out Capability	No
Power supply	PoE on GigE interface, USB in USB interface , 2 DC supply on 5GigE interface
Distance Coverage	Lens dependance

Feature	Specification
Price	€1,139.00
Interface	USB 3.0 or GigE or 5GigE
Network Interface	USB 3.0 Micro-B Pylon SDK or GigE RJ45 or 5GigE RJ45 (model dependent)
Video Streaming Protocol	USB3 Vision / GigE Vision (no RTSP — industrial protocol), Compression Beyond
Control Interface	pylon SDK (C++/C#/Python), pylon Viewer desktop app, GenICam, HW/SW trigger via M8 6-pin GPIO
Band Width	USB 3.0: 5 Gbps, GigE: 1 Gbps, 5GigE: 5 Gbps
Lens Mount	C-Mount

lens for basler ace 2

Best Technical Fit  Basler Lens C11-1620-12M f16mm Details		Best Price/Performance  Basler Lens C23-1620-5M f16mm Details		 Basler Lens C23-1628-16M f16mm Details	
Focal Length	16 mm	Focal Length	16 mm	Focal Length	16 mm
Mount	C-mount	Mount	C-mount	Mount	C-mount
Iris	F2 - F16	Iris	F2 - F16	Iris	F2.8 - F16
min. Working Distance	<u>100 mm</u>	min. Working Distance	<u>200 mm</u>	min. Working Distance	<u>100 mm</u>
List Price	€259.00	List Price	€106.00	List Price	€239.00

Basler ace 2 R Pro(a2A4096-30umBAS)

- use this SDK



Feature	Specification
Resolution	12.3 Megapixels (4096 × 3000)
Shutter Speed	Global Shutter (14µs to 10 seconds)
FPS	29.3 fps at 4096×3000 (default), up to 31.7 fps (throughput limit off)
Field of View (FOV)	Lens dependance
Zoom In/Out Capability	No
Power supply	PoE on GigE interface, USB in USB interface
Distance Coverage	Lens dependance
Price	€1,339.00
Interface	USB 3.0 or GigE
Dimension	42.8 mm × 29 mm × 29 mm
Weight	85g
Network Interface	USB 3.0 Micro-B or GigE RJ45 (model dependent) Pylon SDK
Video Streaming Protocol	USB3 Vision / GigE Vision (no RTSP — industrial protocol), Compression Beyond
Control Interface	pylon SDK (C++/C#/Python), pylon Viewer desktop app, GenICam, HW/SW trigger
Band Width	USB 3.0: 5 Gbps, GigE: 1 Gbps
Lens Mount	C-Mount

Lenses for basler ace 2

- Basler Lens
 - C11-1620-12M f16mm: Φ 40.5 × 61.5 mm
 - C23-1620-5M f16mm: Φ 29.0 × 38.3 mm
 - C23-1628-16M f16mm: Φ 31.8 × 39.5 – 41.5 mm

Best Technical Fit

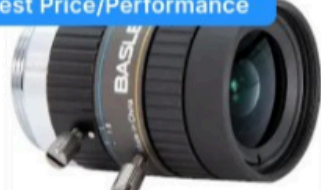


Basler Lens C11-1620-12M f16mm

[Details](#)

Focal Length	16 mm
Mount	C-mount
Iris	F2 - F16
min. Working Distance	<u>100 mm</u>
List Price	€259.00

Best Price/Performance



Basler Lens C23-1620-5M f16mm

[Details](#)

Focal Length	16 mm
Mount	C-mount
Iris	F2 - F16
min. Working Distance	<u>200 mm</u>
List Price	€106.00



Basler Lens C23-1628-16M f16mm

[Details](#)

Focal Length	16 mm
Mount	C-mount
Iris	F2.8 - F16
min. Working Distance	<u>100 mm</u>
List Price	€239.00